Triax Technologies

# S.I.G.P.D.

## Inglés

## TriAxis Technologies

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# Index

**Index** 1

**English video Script** 2

**Abstract** 4

**SWOT Analysis** 5

- **Strengths** 5
- **Weaknesses** 5
- **Opportunities** 6
- **Threats** 7

**English video Script**

We are TriAxis Technologies. We are three workers from the software development and information technology field, that is the reason for choosing this name.

The company is constituted by:

- Santiago Marichal: the lead coordinator and main programmer;
- Santiago da Rosa: the sub-coordinator who is in charge of the frontend; and
- Joaquín Estefan: who is in charge of the backend.

Our mission is to create I.T. systems that improve efficiency and solve real problems for individuals or companies. Our vision is to help those clients get reliable and user-friendly technology that makes their everyday tasks easier and more efficient.

Our first project is the development of two modules for “Hospital de Clínicas” that belong to the ASSE network.

The first module focuses on the digitalization of patient documentation. Currently, many records are only available on paper, which makes the process slow and expensive. Our goal is to create a secure and efficient database system that improves long-term patient management.

The second module is an ambulance tracking system. The system allows the administration to have certain information about the ambulance that is used at the moment. The web application will display the status of the ambulance, with other data, such as who is the driver or the doctor, and more.

One of our main priorities is creating a user-friendly interface because many workers and patients will use the system daily. For us, a successful project means improving hospital efficiency, reducing delays, and creating reliable software for healthcare professionals and users.

Some challenges we may face include database management, system security, and maintaining clear communication during development. We also consider as a challenge, the fact that the software is being created simultaneously with our learning process of certain platforms and technical knowledge.

At TriAxis Technologies, we believe teamwork, organization, and communication are the keys to success.

**Abstract**

*Objective.* With the development of this software, our objective is to make the work of the staff and patients at Hospital de Clínicas easier.

*Reason.* The project was motivated by the need to reduce the use of physical documentation in the hospital and improve the management of ambulance transfers. Currently, these processes can require the use of paper and manual coordination, which can make the staff´s work more demanding.

*Methodology.* We received documentation containing all the software requirements. Based on this documentation, we started working on software engineering documentation as well as on the development of software. Later, we had an interview with some hospital staff members, who suggested several changes to the project. These changes directly affected our development process and required us to change some software´s aspects.

*Result.* As a result, the software is expected to make the staff's work easier by allowing them to focus on other tasks rather than on managing documentation and ambulance transfers. Additionally, the application is designed with a user-friendly interface to facilitate its use and prevent difficulties for employees. Based on the available data, the implementation of the digital system could result in an estimated saving of approximately USD 60,000 in paper usage. However, this estimation only considers the potential savings related to paper and does not include other costs or savings, such as ink or toner, printer maintenance, electricity consumption, staff working hours, training, system maintenance, or development and implementation costs, as sufficient information is not currently available. Therefore, a definitive Return on Investment (ROI) and cost-benefit ratio cannot yet be calculated.

*Conclusion.* These results demonstrate that the proposed software can contribute to the digitalization of hospital processes and reduce the use of physical resources. Overall, the

project provides a practical solution to some needs identified during the initial analysis and interviews.

## SWOT Analysis

### Strengths

- **Knowledge acquired during the first stage:** the group has acquired knowledge in programming, databases, design, documentation, and project management, which can be applied during the next stage.
- **Interdisciplinary teamwork:** the project allowed the group to integrate knowledge from different subjects, combining technical, administrative, business, and research aspects.
- **Project progress:** by the end of the first stage, the group has a foundation of documentation, requirements, design, and development that allows them to continue working on the system.
- **Distribution of responsibilities:** the members have defined roles and can divide tasks according to the needs of the project.
- **Contact with a real institution:** working on needs identified by Hospital de Clínicas allowed the group to bring the academic project closer to a real-world situation.

### Weaknesses

- **Lack of experience with projects of this scale:** some members may have limited previous experience developing a system involving multiple modules and different technologies.
- **Organizational difficulties:** coordinating programming, documentation, research, and business-related tasks can make time management and task distribution more difficult.

- **Dependence on teamwork:** if one or more members are unable to complete their assigned tasks, other parts of the project may be delayed.
- **Need to improve internal communication:** changes in the requirements can cause confusion if they are not properly communicated and documented within the group.
- **Changes in requirements:** during the first stage, interviews with hospital staff led to modifications to the project, which may require reorganizing some of the work already completed.
- **Limited time:** the group must develop the software while also completing assignments and activities from different subjects.

**Opportunities**

- **Continue improving the software:** the second stage provides an opportunity to implement new functionalities, fix errors, and improve the modules already developed.
- **Apply the knowledge acquired:** the knowledge gained during the first stage can be used to improve both the development and the documentation of the project.
- **Conduct further testing:** having a more advanced version of the system will allow the group to identify problems that were not detected during the initial stages.
- **Integration between modules:** there is an opportunity to improve the relationship between the different parts of the system and create a more complete solution.
- **Improve the project's presentation:** the second stage provides an opportunity to improve the documentation, user interface, and final presentation of the project.
- **Turn the academic project into an applicable solution:** contact with a real institution may allow the project to develop beyond being a simple academic assignment and become a solution that could potentially be applied in a real environment.

**Threats**

- **New changes in requirements:** hospital staff may identify new needs that require modifications to functionalities that have already been developed.
- **Time limitations:** the number of assignments and deadlines during the second stage could make it difficult to dedicate enough time to the project.
- **Technical problems:** software errors or problems with the server, database, or development tools could delay the development process.
- **Difficulties obtaining information:** if the group is unable to maintain communication with hospital staff, it could become more difficult to validate decisions made during the development process.
- **Technological changes:** the technologies used in the project may require updates or modifications during development.
- **Difficulty implementing the solution in a real environment:** although the system may work correctly in an academic environment, additional requirements may arise when adapting it to the hospital's real conditions.
- **Academic workload from different subjects:** the simultaneous demands of different subjects may affect the group's availability to make progress on the project.